

Sukrityranjan Samanta, Ph.D. student

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Education

Virginia Tech

Aug 2023 - Present

Ph.D. in Geotechnical engineering (GPA: 3.64/4.00)

Blacksburg, Virginia

- **Technical Coursework:** Soil behavior, Foundation engineering I & II, Geotech earthquake engineering, Numerical methods, Seepage & earth structures, Risk analysis in Geotech Engr, Geological engineering, ML application in CEE, Soil and site improvements, and Statistics in research.
- **Future Professoriate Graduate Certificate:** Preparing future professoriate, Contemporary pedagogy, and Communicating science.

National Taiwan University

Apr 2021 - Jun 2023

Masters of Science in Geotechnical engineering (GPA: 3.95/4.30)

Taipei, Taiwan

National Institute of Technology

Jul 2013 - Jun 2017

Bachelors of Technology in Civil engineering (GPA: 8.41/10.00)

Raipur, India

Experience

Virginia Tech

Aug 2023 - Present

Graduate research assistant

Blacksburg, Virginia

- ERDC Project: Numerical analysis of progression and collapse of underground cavities in levees (Jan'26 - Present).
- VTRC project: Design Adequacy Analysis of the Poplar Creek Box Culverts (May'25 - Dec'25).
- NSF project: Transformation Understanding of Rainfall-Triggered Landslides from a Climate Change Perspective: Initiation and Consequences (Aug'24 - Apr'25).
- CGPR project: Controlling Settlement Due to Adjacent Loading Using Sheet-Pile Walls: Insights from Finite Element Analysis. (Aug'23 - Jul'24).

National Taiwan University

Apr 2021 - Jun 2021

Graduate research assistant

Taipei, Taiwan

- Conducted model tests on geosynthetic encased column (GEC)-stabilized slopes subjected to seepage.
- Performed fully coupled flow-deformation finite element analyses and validated the hydro-mechanical behavior.
- Designed and instrumented narrow geosynthetic reinforced soil (NGRS) structures with sensors to monitor deformation, pore water pressure, and long-term field performance.

KEC International Limited

Jul 2017 - Feb 2021

Project Engineer

Kabul, Afghanistan

- Project: 220kV transmission line from Ghazni to Kandahar, and 500kV transmission line CASA-1000 Lot-2.
- Prepared FIDIC documents for competitive bid preparation, financial modeling of projects, operating with baseline schedule.
- Coordinated with the client for any technical issue resolution and deal with a cross-functional decision-making team for the successful completion of multi-million-dollar projects.

Professional Service During Ph.D.

Education Chair, Geotechnical Engineering Student Organization

Jan 2024 - Dec 2024

- Organized student engagement activities for graduate students in geotechnical engineering.
- Participated in ASCE outreach events and school visits to demonstrate geotechnical engineering concepts and hands-on activities to K-12 students.

Publications During Ph.D.

- **Samanta, S.,** Hosseini, R., and Yerro, A. (2026). Controlling Settlement Due to Adjacent Loading Using Sheet-Pile Walls: Insights from Finite Element Analysis. *Geotechnical and Geological Engineering*, 44(1), 44.
- Karam, C., Yerro, A., and **Samanta, S.** (2026). Studying Underground Cavity Progression and the Effects under Levees: A Numerical Investigation Using the Material Point Method. In *Proceedings of Geo-Congress 2026* (pp. 229-239).
- **Samanta, S.,** Yerro, A., and Dhakal, S. (2025). A Data-Driven Approach to Predicting the Stability of Root-Reinforced Slopes under Extreme Rainfall Conditions. In *Proceedings of Geo-Extreme 2025* (pp. 182-193).
- **Samanta, S.,** Yerro, A., and Hosseini, R. (2025). A Numerical Study on the Effectiveness of Driven Sheet-Pile Walls as Barriers to Reduce Settlement in Preexisting Structures due to Adjacent Loading. In *Proceedings of Geotechnical Frontiers 2025* (pp. 163-173).

Technical Skills

Languages: MATLAB, Python, FORTRAN.

Technologies: Auto CAD, Staad.Pro, PTV.Vissim, PLAXIS- 2D/3D, SAP 2000, Settle3D, SLIDE2D, Geostudio package, Rocscience package, Metashape.

Licenses: Chartered Engineer - "The Institution of Engineer (India)", OSHA-30 Hour Construction Safety Certification, Associate Professional Engineer - "Engineering Council of India", NCEES – Fundamentals of Engineering (FE) and Principles & Practice of Engineering (PE) Examinations Passed; Licensure pending due to degree requirements..